SCaLAR@NITK at HalluScoring-2026: A Reference-Free Generative Approach to Hallucination Detection via Teacher-Student Distillation

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Abstract

Large language models (LLMs) produce fluent but factually incorrect outputs which become significant when such models are applied to local languages, such as Arabic, because of the language's complex morphology, dialectal richness, and the insufficient material available for training and evaluation. The competition HalluScoring-2026 Track 1 aims to address these problems and consists of two tasks that demand hallucination detection in Arabic question-answering settings. To solve this issue, we transfer the knowledge of a powerful reference-based teacher that is a stacked ensemble of five fine-tuned encoder classifiers with dynamic lexical, embedding, and entailment characteristics onto a fine-tuned, open-weight generative student model whose input is restricted to exactly the fields available at test time.

1 Introduction

Although large language models (LLMs) are deployed for Arabic Q&A-type tasks, determining when an LLM's answer is fictitious or hallucinated remains challenging, as there are fewer tools for detecting hallucinations in Arabic than in English. The HalluScoring-2026 shared task (Alansari et al., 2026) addresses this directly: Track 1 (Tasks 1.1 and 1.2) poses hallucination detection as binary classification, where Task 1.1 evaluates generalization to unseen questions generated by the same set of LLMs, whereas Task 1.2 evaluates cross-model generalization by testing on hallucinations produced by previously unseen LLMs.

Our main system strategy is a two-stage teacher-student pipeline. First, we train a reference-based teacher, which is an ensemble of five fine-tuned encoder classifiers combined via a stacked meta-model with lexical, embedding-similarity, and natural language inference (NLI) features computed against the gold reference answer. Second, because the released test set omits the gold reference entirely (only the question and generated answer are provided), we distill the teacher's judgments into a fine-tuned generative student model whose input is restricted to exactly the fields the test set actually provides. Hence, the submitted system never depends on information it will not have at the time of evaluation.

2 Background

Task setup: Tasks 1.1 and 1.2 take as input a tuple (question, gold answer, generator model, generated answer) and require a binary hallucination label plus a continuous $P(\text{hallucinated}) \in [0, 1]$ as output. For example, consider the question *"What is thermal expansion?"* The gold answer states that it is *"an increase in the dimensions of a material as a result of an increase in its temperature."* In contrast, the generated answer states that thermal expansion is *"a decrease in the dimensions of a material when its temperature increases due to the contraction of molecules."* Because the generated answer directly contradicts the gold answer, its correct label is 1 (hallucinated). Critically, the released test set provides only the question and the generated answer with no gold reference and no label, which we treat as a hard constraint on any submitted system's inference-time inputs.

Dataset: The HalluScore dataset (Alansari and Luqman, 2026), used in Track1 of the HalluScoring 2026 shared task, consists of Arabic question–answer instances paired with LLM-generated responses annotated with binary hallucination labels. Both tasks 1.1 and 1.2 take place in a shared hallucination detection environment; however, they have different aims with respect to generalization. The training dataset consists of 4,705 samples with 2,221 questions. Task 1.2's dev dataset consists of samples generated by two hidden models. Therefore, task 1.2 represents a stricter probe of a de-

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Table 1: Dataset statistics for HalluScoring 2026.

Dataset	Split	Instances	Questions	Models	Model Type
Task1.1	Train	4,705	2,221	5	Seen
	Dev	1300	900	5	Seen
	Test	1030	206	5	Seen
Task1.2	Train	4,705	2,221	5	Seen
	Development	200	100	2	Unseen
	Test	412	206	2	Unseen

tector's performance in perceiving hallucination signals that do not depend on the generators rather than on the effects produced by the models present during training. The dataset distribution is as seen in the Table 1

Tracks: We participated in Track 1, submitting to both Task 1.1 and Task 1.2, using a single underlying system architecture (§3) trained separately per task on each task's respective data split.

Related work. Hallucination in machine-generated text is well-studied broadly (Saxena and Bhattacharyya, 2024), which categorizes hallucinations as intrinsic, extrinsic, factual, or faithfulness errors, but Arabic-specific evidence suggests the problem is both more severe and less studied: Sibaee et al. (2024) find that Arabic's morphological complexity compounds hallucination detection difficulty, and Mubarak et al. (2024) show empirically that roughly a quarter of ChatGPT/GPT-4-generated Arabic sentences in their Halwasa dataset are factually incorrect, underscoring the need for detection methods built for Arabic specifically, rather than transferred from English tooling.

The dominant approach to this problem is reference-based: comparing a candidate answer against a trusted reference via NLI-style entailment checking, lexical overlap, and embedding similarity (R, 2025). However, Kang et al. (2024)'s multilingual comparison shows this approach's central weakness for our setting: NLI-based metrics perform well for sentence-level hallucinations in high-resource languages but degrade on atomic facts and lower-resource languages where the regime Arabic falls into. More fundamentally, every reference-based method in this line of work assumes a trusted reference is available at evaluation time, an assumption that HalluScoring-2026's test set does not satisfy. We address this gap directly: rather than abandoning the strength of reference-based detection, we retain it during training and transfer its judgments through the knowledge distillation framework Hinton et al. (2015) into a student that operates without a reference at inference time.

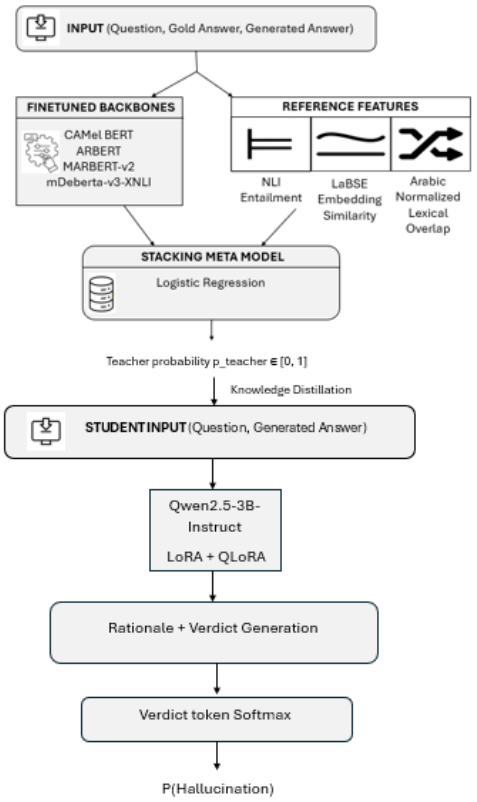


Figure 1: Overview of the Student-Teacher Model.

3 System Overview

3.1 Reference-Based Teacher Ensemble

We fine-tune five encoder classifiers on (question, gold answer, generated answer) triples: CAMeL-BERT (Inoue et al., 2022), ARBERT and MARBERTv2 (Abdul-Mageed et al., 2021), multilingual BERT, and an mDeBERTa-v3 NLI model. Out-of-fold (5-fold) predictions on the training split provide leakage-free inputs to a stacking stage. Along with these five backbone probabilities, we compute reference-based engineered features per instance such as NLI entailment/contradiction scores, LaBSE embedding cosine similarity, lexical (Jaccard) overlap, and named-entity/numeric mismatch indicators between the gold and generated answers. A logistic regression meta-classifier combines all of these signals and is validated with repeated stratified cross-validation. This teacher requires the gold reference answer and is therefore never itself submitted for test-set prediction. The overview of this student-teacher ensemble is as shown in the Figure 1.

Algorithm 1 Teacher-to-student distillation

Require: Training set $\mathcal{D} = \{(q_i, g_i, a_i, y_i)\}_{i=1}^{N}$; teacher T; features F_i; student S_θ; KL weight λ

$\mathcal{D}' \leftarrow \emptyset$
for each $(q_i, g_i, a_i, y_i) \in \mathcal{D}$ **do**
  $p_i \leftarrow T(q_i, g_i, a_i, F_i)$
  $\hat{y}_i \leftarrow 1[p_i \geq 0.5]$
  $r_i \leftarrow \textsc{Rationale}(g_i, a_i, F_i, \hat{y}_i)$
  $t_i \leftarrow r_i \,\|\, \textsc{Verdict}(\hat{y}_i)$
  $x_i \leftarrow \textsc{Prompt}(q_i, a_i)$
  $\mathcal{D}' \leftarrow \mathcal{D}' \cup \{(x_i, t_i, p_i)\}$
end for
for each minibatch $\{(x_i, t_i, p_i)\}_{i=1}^{B} \subset \mathcal{D}'$ **do**
  $s_i \leftarrow \textsc{Softmax}(S_\theta(x_i))$ over HALLUC. and NOT HALLUC.
  $\mathcal{L}_{\text{SFT}} \leftarrow \frac{1}{B}\sum_i \textsc{CrossEntropy}(S_\theta, x_i, t_i)$
  $\mathcal{L}_{\text{KD}} \leftarrow \frac{1}{B}\sum_i D_{\text{KL}}((p_i, 1-p_i) \,\|\, s_i)$
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{SFT}} + \lambda \mathcal{L}_{\text{KD}}$
  Update θ using $\nabla_\theta \mathcal{L}$
end for

3.2 Distillation into a Reference-Free Student

The main challenge is to transfer the teacher's discriminative signal without transferring its dependence on the gold reference. For each training instance, we obtain the teacher's calibrated probability p_{teacher} and construct a supervised fine-tuning example whose *input* contains only the question and generated answer, and whose *target* is a short rationale stating the correct fact directly (e.g., "the correct answer is Canberra; the generated answer states something different") followed by a verdict token, rather than a rationale that references "the gold answer" as a field the student will not have.

We fine-tune Qwen2.5-3B-Instruct, an open-weight decoder under the shared task's 13B-parameter limit, using LoRA (Hu et al., 2025) with 4-bit quantization (QLoRA; Dettmers et al., 2023). The training loss combines a standard supervised fine-tuning term $\mathcal{L}_{\text{SFT}}$ with an auxiliary knowledge-distillation term $\mathcal{L}_{\text{KD}}$: a KL-divergence between the teacher's Bernoulli verdict distribution $(p_{\text{teacher}}, 1-p_{\text{teacher}})$ and the student's own predicted distribution over the "Hallucinated" vs. "Not Hallucinated" token at the position immediately following the prompt,

$$\mathcal{L} = \mathcal{L}_{\text{SFT}} + \lambda \cdot \mathcal{L}_{\text{KD}} \tag{1}$$

following the general distillation framework of Hinton et al. (2015), with λ controlling the weight of the calibration term relative to the standard fine-tuning loss. Algorithm 1 shows the full teacher-to-student distillation procedure.

During inference, the student model has access only to the question and the output answer. For each instance, we obtain its score, which is the softmax over just the two verdict tokens in the first position, allowing us to compute the AUC-ROC continuously. Additionally, we decode the full rationale for qualitative evaluation. No NLI or embedding or entity matching computations are performed at inference, and only the fine-tuned generative model is run.

We submit a single system configuration to both Task 1.1 and Task 1.2, trained separately on each task's training/development split, with identical architecture.

Table 2: Official leaderboard results (%) for our submitted teacher and student systems on Tasks 1.1 and 1.2.

Task	System	Split	AUC-ROC	F1-M	AUC-PR
1.1	Teacher	Dev	97.91	94.66	98.57
1.1	Student	Dev	81.48	78.35	82.73
1.1	Student	Test	64.92	62.98	86.97
1.2	Teacher	Dev	99.58	96.28	99.75
1.2	Student	Dev	60.86	64.38	68.63
1.2	Student	Test	67.97	67.21	86.37

4 Experimental Setup

Data splits. Backbones and the stacker are trained on each task's training split with out-of-fold predictions used for stacking. The student is fine-tuned on distillation data derived from the training split only.

Hyperparameters. The proposed model with full input encoding (question + gold answer + generated answer), default training epochs per backbone, and 5-fold out-of-fold prediction generation for stacking. Stacker uses a logistic regression meta-classifier, 5-fold $\times$ 5-repeat stratified cross-validation, and SMOTE oversampling enabled for Task 1.2 (smaller, more imbalanced dev split) but not for Task 1.1. The student model is a Qwen2.5-3B-Instruct base model, LoRA rank R=16, α=32, dropout 0.05, targeting all attention and MLP projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`), learning rate 2e-4, effective batch size 16 (per-device batch size 4 $\times$ gradient accumulation 4), 3 epochs, KL weight λ=0.5, max sequence length 768, 4-bit (QLoRA) quantization with non-reentrant gradient checkpointing, and single-GPU training.

5 Results

Quantitative findings. Table 2 reports our official leaderboard results. A striking pattern emerges across the two tasks: Task 1.1 AUC-ROC *drops* from 81.48 (dev) to 64.92 (test), a 16.6-point fall, while Task 1.2 AUC-ROC *rises* from 60.86 (dev) to 67.97 (test), a 7.1-point gain, the opposite direction. Since both tasks share the same training data and system configuration, this asymmetry is attributable to the composition of each task's unseen-generator dev/test splits rather than to the model itself. Task 1.2's dev split is small (200 instances, two anonymized generators) and imbalanced, so its dev AUC-ROC is a noisier estimate of true generalization than Task 1.1's; it is plausible that Task 1.2's particular dev-set generator pairing happened to be unusually difficult relative to its test-set pairing, while Task 1.1's dev pairing happened to be unusually easy relative to its test pairing. For the student, AUC-PR ranges from 68.63 to 86.97 across the reported development and test splits, suggesting the system retains strong precision-orientated ranking quality even where AUC-ROC fluctuates.

6 Conclusion and Future Work

We have developed a teacher-student architecture for the HalluScoring 2026 Tasks 1.1 and 1.2 that converts a powerful ensemble classifier that uses references into a generative model that does not rely on references. This demonstrates that task-defined reliability at the development/test scale splits input tasks and is as large a factor in overall performance as the gap between reference-based and reference-free architectures, since cross-model evaluation is sensitive to which unseen generators are included in the split. Nevertheless, the main limitation of the solution is that the student's knowledge (which has 3 billion parameters) cannot surpass the knowledge of the references used during distillation. Future research will focus on employing larger student backbones, removing the KL term, and conducting error analyses on official testing, with other pairs of generators held aside during the experiment.

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